This assignment will not be graded and is only for practice.

Functions of several variables:

Level 1:

1) Which of the following are correct with respect to $f(x, y) = x^2 - 2y$, where x, y are real? ***1 point***

- [ ] Domain of f is $\mathbb{R}^2$
- [ ] Domain of f is $\mathbb{R}$
- [ ] Range of f is $\mathbb{R}$
- [ ] Range of f is $\mathbb{R}^2$

No, the answer is incorrect.
Score: 0

Accepted Answers:

Domain of f is $\mathbb{R}^2$
Range of f is $\mathbb{R}$

2) Let $D \subset \mathbb{R}^n$ and $f : D \longrightarrow \mathbb{R}^m, g : D \longrightarrow \mathbb{R}^m$ be multivariable functions on D. Which of the following statements is(are) TRUE? ***1 point***

- [ ] The product function fg is defined on D by $fg(\underset{\sim}{x}) = f(\underset{\sim}{x}) \times g(\underset{\sim}{x}), \underset{\sim}{x} \in D$
- [ ] The sum function $f + g$ is defined on D by $(f + g)(\underset{\sim}{x}) = f(\underset{\sim}{x}) + g(\underset{\sim}{x}), \underset{\sim}{x} \in D$
- [ ] Let $c \in \mathbb{R}$. The function cf is defined on D by $(cf)(\underset{\sim}{x}) = \frac{1}{c} \times f(\underset{\sim}{x}), \underset{\sim}{x} \in D$
- [ ] If $m = 1$ and $g(\underset{\sim}{x}) = 0, \underset{\sim}{x} \in D$, then the function f/g is defined on D by $(f/g)(\underset{\sim}{x}) = f(\underset{\sim}{x})/g(\underset{\sim}{x}), \underset{\sim}{x} \in D$

No, the answer is incorrect.
Score: 0

Accepted Answers:

The sum function $f+g$ is defined on D by $(f+g)(\underset{\sim}{x}) = f(\underset{\sim}{x}) + g(\underset{\sim}{x}),\ \underset{\sim}{x} \in D$

If $m = 1$ and $g(\underset{\sim}{x}) = 0,\ \underset{\sim}{x} \in D$, then the function f/g is defined on D by $(f/g)(\underset{\sim}{x}) = f(\underset{\sim}{x})/g(\underset{\sim}{x}),\ \underset{\sim}{x} \in D$

3) Suppose $f : D \longrightarrow \mathrm{R}$ is a function defined on a domain $D \subset \mathrm{R}$. Which of the following can be f? **1 point**

- ☐ $f(x) = 4 \pm \sqrt{x-10}$
- ☐ $f(x) = \frac{x^2}{x+1}$
- ☐ $f(x) = log(\sqrt{x^4+3})$
- ☐ $f(x) = x^3 + x^2 - x$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$f(x) = \frac{x^2}{x+1}$
$f(x) = log(\sqrt{x^4+3})$
$f(x) = x^3 + x^2 - x$

4) Which of the following statements is(are) TRUE? **1 point**

- ☐ The output of a scalar valued multivariable function is always a real number.
- ☐ The output of a vector valued multivariable function can be a real number.
- ☐ The output of a single variable vector valued function is always a vector.
- ☐ The output of a vector valued multivariable function can not be a real number.

No, the answer is incorrect.
Score: 0

Accepted Answers:

The output of a scalar valued multivariable function is always a real number.
The output of a single variable vector valued function is always a vector.
The output of a vector valued multivariable function can not be a real number.

5) Consider the function $S(t) = (\cos t, \sin t, 2t)$, where t is real. Choose the set of correct options. **1 point**

- [ ] S is a single-variable vector-valued function
- [ ] S is a multi-variable vector-valued function
- [ ] The domain of S is R and the co-domain of S is R^3
- [ ] The domain of S is R^3 and the co-domain of S is R^3

No, the answer is incorrect.

Score: 0

Accepted Answers:

S is a single-variable vector-valued function
The domain of S is R and the co-domain of S is R^3

6) How many of the following functions is(are) a vector valued multivariable function?

- $f(x, y) = \frac{1}{3\pi} e^{x^2+y^2}$
- $f(x, y) = (x^3, y^2)$
- $f(x, y) = (x + y, x - y, 2xy)$
- $f(x, y, z) = x^2 + xyz - z^2$
- $f(x) = (x + 1, x - 1)$
- $f(x, y, z) = (e^{x+y+z}, 0, 0)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

7) Suppose $g : D \longrightarrow \mathrm{R}^3$ is a function defined on a domain $D \subset \mathrm{R}^2$. Which of the following functions may be g? **1 point**

- [ ] $g(x, y) = 4x + 3y - 12xy$
- [ ] $g(x, y) = (x^4 + 1, y^4 + 1, xy)$
- [ ] $g(x, y, z) = (e^x - e^z, e^z - e^y)$

- [ ] $g(x, y) = (sin(x + y), cos(x - y))$

- [ ] $g(x, y) = (0, sin(e^x), cos(e^x))$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$g(x, y) = (x^4 + 1, y^4 + 1, xy)$
$g(x, y) = (0, sin(e^x), cos(e^x))$

Level 2:

8) Which of the following are correct with respect to $f(x, y) = \sqrt{1 - \frac{x^2}{9} - \frac{y^2}{16}}$, where x, y are real? *1 point*

- [ ] Domain of f is $D = \{(x, y) \mid \frac{x^2}{9} + \frac{y^2}{16} \leq 1\}$

- [ ] Domain of f is $D = \{(x, y) \mid \frac{x^2}{9} + \frac{y^2}{16} \geq 1\}$

- [ ] Range of f is $[0, 1]$

- [ ] Range of f is $[0, 1)$

- [ ] Range of f is $[1, \infty)$

No, the answer is incorrect.
Score: 0

Accepted Answers:

Domain of f is $D = \{(x, y) \mid \frac{x^2}{9} + \frac{y^2}{16} \leq 1\}$
Range of f is $[0, 1]$

9) A function $F : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ has the form $F(x, y) = (P(x, y), Q(x, y), R(x, y))$. Choose the correct option(s). *1 point*

- [ ] F is a multi-variable vector-valued function

- [ ] F is a single-variable vector-valued function
- [ ] P, Q, R are single-variable vector-valued functions
- [ ] P, Q, R are multi-variable vector-valued functions
- [ ] P, Q, R are multi-variable scalar-valued functions

No, the answer is incorrect.

Score: 0

Accepted Answers:

F is a multi-variable vector-valued function
P, Q, R are multi-variable scalar-valued functions

10) Consider $f(x, y) = \sqrt{1 - \frac{x^2}{9} - \frac{y^2}{4}}$. Note: An ellipse is of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where a, b are the dimensions of the ellipse. ***1 point***

- [ ] Ellipses of increasing size represent the curves $\{(x, y) | f(x, y) = c\}$, where c increases from 0 towards 1
- [ ] Ellipses of decreasing size represent the curves $\{(x, y) | f(x, y) = c\}$, where c increases from 0 towards 1
- [ ] Ellipses of constant size represent the curves $\{(x, y) | f(x, y) = c\}$, where c increases from 0 towards 1
- [ ] The curves $\{(x, y) | f(x, y) = c\}$, where $c \in [0, 1)$, cannot be represented by an ellipse
- [ ] $f(x, y) = 1$ is a point

No, the answer is incorrect.

Score: 0

Accepted Answers:

Ellipses of decreasing size represent the curves $\{(x, y) | f(x, y) = c\}$, where c increases from 0 towards 1
$f(x, y) = 1$ is a point

11) ***1 point***

Consider a point source $S = (1, 2, 3)$ radiating energy. The intensity I at a given point $P = (x, y, z)$ in space is inversely related to the square of the distance d between S and $P : I(x, y, z) = \frac{k}{d^2}$, where k is a real positive constant. Choose the correct option(s).

- [ ] Intensity I is constant on $\{(x, y, z) | (x-1)^2 + (y-2)^2 + (z-3)^2 = 1\}$
- [ ] Intensity I is constant on $\{(x, y, z) | (x-1)^2 + (y-2)^2 + (z-3)^2 = 2\}$
- [ ] Intensity I is constant on $\{(x, y, z) | x^2 + y^2 + z^2 = 1\}$
- [ ] Intensity I is constant on $\{(x, y, z) | x^2 + y^2 + z^2 = 2\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

Intensity I is constant on $\{(x, y, z) | (x-1)^2 + (y-2)^2 + (z-3)^2 = 1\}$
Intensity I is constant on $\{(x, y, z) | (x-1)^2 + (y-2)^2 + (z-3)^2 = 2\}$

Your score is: 0/11